

## LECTURE 2: TRAINING STRATEGY

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## Training the model

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# SETTING HYPERPARAMETERS

**Idea #1:** choose the hyperparameters that work best on *the data*.

**BAD:** the network can memorize the training set.



**Idea #2:** split into **train** and **test**; choose what works best on test.

**BAD:** no idea how it will perform on new data.



**Idea #3:** split into **train**, **val** and **test**; tune on val, evaluate once on test.

**Better!**



tune here

open once, at the end

A typical split is 60 / 20 / 20. The exact numbers matter far less than never tuning on test.

## SETTING HYPERPARAMETERS: CROSS-VALIDATION

### Your dataset

**Idea #4: cross-validation.** Split the data into **folds**, try each fold as validation, and average the results.

|        |        |        |        |        |      |
|--------|--------|--------|--------|--------|------|
| fold 1 | fold 2 | fold 3 | fold 4 | fold 5 | test |
| fold 1 | fold 2 | fold 3 | fold 4 | fold 5 | test |
| fold 1 | fold 2 | fold 3 | fold 4 | fold 5 | test |

Widly used in small datasets

### The medical mistake that voids a study

Split by **patient**, never by image. Folds included.

Two slices of the same tumor in train and test do not measure generalization; they measure memory. This inflates accuracy quietly and is the single most common flaw in published medical deep learning.

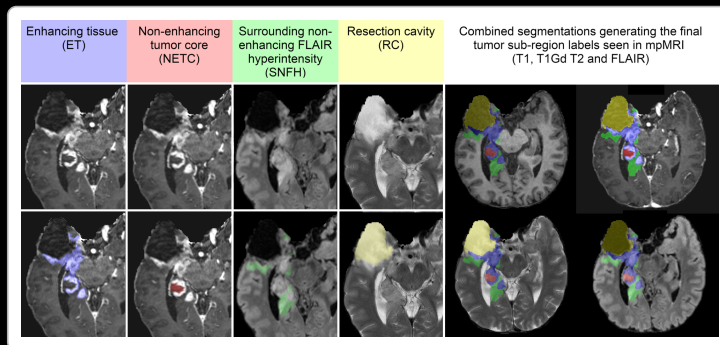
**When labels are scarce**

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## LABELS ARE HARD TO OBTAIN

### Expensive

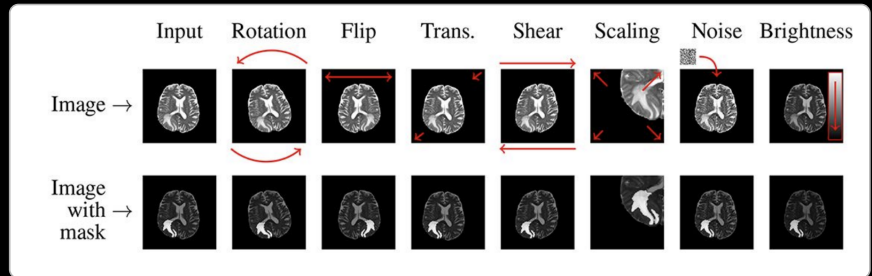
A label needs a **specialist's** time. The 2024 BraTS post-treatment glioma challenge pooled  $\sim 2,200$  cases from seven hospitals in two countries, and the scans were the cheap half.



Each case: four MRI sequences, four tumour sub-regions, **drawn voxel by voxel** in 3D: refined by a clinical annotator, then signed off by a board-certified neuroradiologist, and returned until it passed.

## STRATEGY 1: MAKE YOUR DATA GO FURTHER, WITH AUGMENTATION

Apply affine and pixel transformations to your data → more training samples.



### Why it works

Each transform is a statement about what the label is *invariant* to. You are handing the network a prior for free.

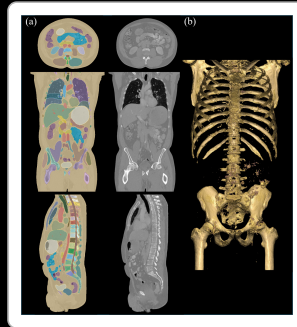
### Note the second row

A geometric transform must be applied to the **mask** as well. Get this wrong and you train on silently mislabeled data.

**Can we automate this?**



## STRATEGY 2: GENERATE THEM, WITH SYNTHETIC IMAGES



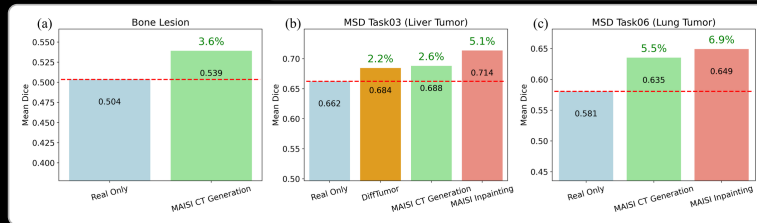
The conditioning organ mask, and the CT it produced.

### MAISI

A **latent diffusion** model for 3D CT. It denoises in a compressed latent space, and a ControlNet conditions the output on an organ mask of 127 anatomical structures, volumes up to  $512 \times 512 \times 768$ .

### More than augmentation

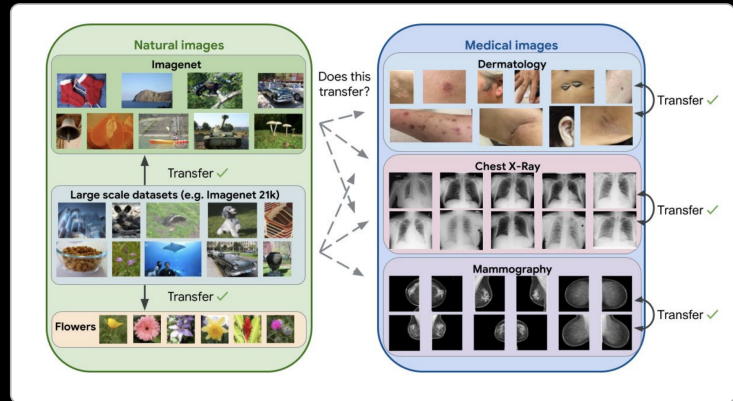
You hand it the mask, so **the label is generated with the image**: a new labelled example, not a re-photographed one. The catch: synthetic anatomy can be confidently wrong. Fine for training, **never** for diagnosis.



Generated volumes mixed into real training data lift tumour segmentation Dice on every task tried; green is the gain over real data alone (three of six panels). We build one properly in Lecture 3.

## STRATEGY 3: BORROW FROM ANOTHER DOMAIN, WITH TRANSFER LEARNING

Pretrain on a large natural-image dataset, then fine-tune on the medical task. **Does that transfer?**



Largely yes. Early filters (edges, textures, gradients) are not specific to cats, and dermatology, chest x-ray and mammography all benefit. Later layers are where the domains genuinely diverge, which is why we fine-tune those.

# HOW MUCH TO FREEZE, HOW MUCH TO FINE-TUNE

|                     |                | similar to ImageNet                                                 | very different                                                           |
|---------------------|----------------|---------------------------------------------------------------------|--------------------------------------------------------------------------|
| very little data    | little         | Freeze the backbone; train a linear classifier on the top features. | Hardest case. Try a linear classifier on <i>earlier</i> -layer features. |
| quite a lot of data | quite a lot of | Fine-tune the last few stages.                                      | Fine-tune most or all of the network.                                    |

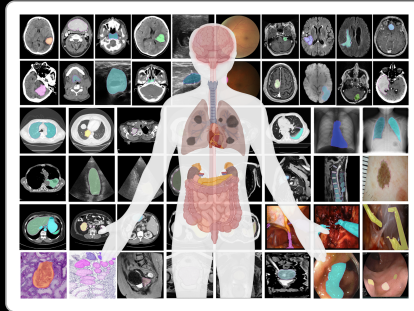
## Use a smaller LR

When fine-tuning, start around **1/10** of the original learning rate. The pretrained weights are already good: large steps destroy them.

## In practice

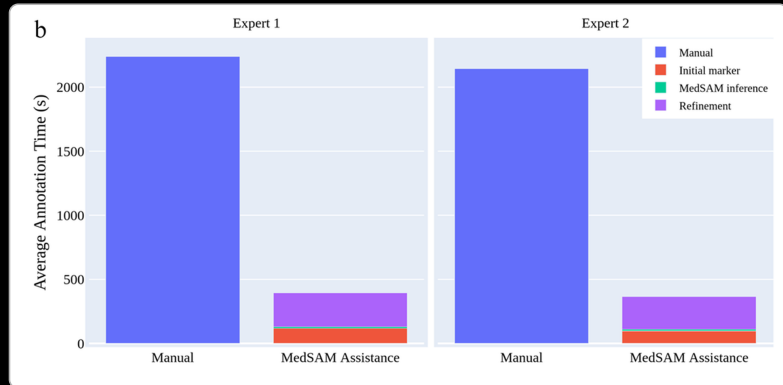
Try **fine-tuning everything** first. It is usually the strongest option and the table above is a fallback for when it overfits.

## NEED AN OUT-OF-THE-BOX TOOL?



One model, any modality.  
The prompt is a **bounding box**.

**1.57M** image-mask pairs  
**10** modalities  
**30+** cancer types



Expert annotation time, **82% less**.